Spatial SIR Cellular Automata: Computational Exploration and Epidemiological Applications

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ABSTRACT

Spatial structure strongly shapes infectious disease dynamics, yet many classical compartmental models assume well-mixed contact. Here we implement a two-dimensional cellular automaton (CA) for an influenza-like illness using a stochastic SIR(S) framework, where each individual occupies a lattice site and interacts locally with its Moore neighborhood. The model couples transmission probability to the number of infected neighbors and includes recovery, mortality, and optional waning immunity. Using a lightweight NumPy and Matplotlib implementation, the code enables interactive experimentation with outbreak thresholds, clustering, and the emergence of spatial wavefronts. This approach is relevant for teaching epidemiological intuition, prototyping interventions (e.g., reduced contact, targeted isolation), and exploring how heterogeneous mixing can alter epidemic curves and final size compared with mean-field assumptions.

Introduction

Infectious diseases propagate through networks of human contacts that are inherently heterogeneous in space and time. While ordinary differential equation (ODE) compartmental models such as Susceptible-Infected-Recovered (SIR) offer analytical tractability, their well-mixed assumption often obscures phenomena driven by locality, such as clustered transmission, spatial wave propagation, and the persistence of infection in pockets of susceptibility. Cellular automata provide an alternative computational lens: individuals are represented explicitly on a grid, and epidemic spread emerges from repeated application of simple, local interaction rules.

A key quantity in epidemiology is the (effective) reproduction number, which summarizes how many secondary infections arise from a typical infectious individual. In well-mixed models, this quantity depends primarily on average contact rates and infectious duration. In spatial models, however, the same average parameters can yield very different effective reproduction because infected individuals repeatedly contact the same neighbors, and because susceptible pathways can be interrupted by recovered or isolated individuals. Consequently, spatial models can display slower-than-expected growth, local extinctions, and strong sensitivity to initial seeding location.

Influenza-like illness is a suitable example for this type of modeling because it combines fast generation times with strong seasonality and because interventions often act by reducing local exposure (e.g., improved ventilation in specific settings) rather than changing population-wide mixing instantly. The CA framework makes these localized mechanisms explicit: infection spreads as connected clusters on the lattice, and control measures act by breaking or weakening these connections.

This work adapts the style and section structure of a prior computational exploration of Conway's Game of Life to an epidemiological setting, maintaining a concise scientific-report layout while shifting the underlying rules from birth-and-death pattern formation to transmission-and-recovery dynamics.

We focus on an influenza-like illness (ILI) as a representative acute respiratory infection with short infectious periods, substantial person-to-person transmission, and context-dependent mortality risk. The objective is not to fit a specific historical outbreak, but to provide a transparent simulator where users can test how local contact intensity (transmissibility), clinical course (recovery), and severity (mortality) shape emergent outbreak patterns. By implementing the model in Python with real-time animation, the code serves as a practical bridge between theoretical epidemiology and computational experimentation.

Historical Context

Epidemic modeling has a long tradition of balancing realism and mathematical simplicity. Early compartmental approaches formalized the idea that populations can be partitioned into health states and that transitions between states determine incidence and prevalence. However, these models typically treat contacts as uniformly random. Over time, researchers introduced spatially explicit methods - including metapopulation models, partial differential equations, network models, and cellular automata - to capture the effects of locality and heterogeneous mixing.

Cellular automata, originally developed in the context of self-reproducing machines and complex systems, are particularly appealing for epidemiology because they emphasize local interactions, synchronous updating, and emergent behavior. Instead of prescribing a global epidemic curve, CA models allow outbreaks to arise from micro-level rules applied to each individual at each time step. This perspective is valuable when transmission depends on household proximity, neighborhood density, workplace clustering, or mobility constraints.

In practice, CA epidemic models have been used as pedagogical tools and as exploratory simulators for non-pharmaceutical interventions. Simple rule changes - such as reducing effective contacts, introducing movement restrictions, or isolating symptomatic individuals - can be expressed naturally in the automaton. The resulting spatiotemporal patterns help build intuition about why the same pathogen can produce different outcomes across communities with different contact structures.

Lattice-based epidemic models can also be interpreted through the lens of percolation and contact processes, where epidemic takeoff requires a sufficiently connected set of susceptible sites for the infection to traverse. This viewpoint naturally explains why small changes in transmissibility or susceptible density can trigger sharp transitions between minor outbreaks and large-scale spread. In the CA setting, such transitions are visible as the difference between short-lived clusters that burn out and long-lived wavefronts that traverse the grid.

Modern computational epidemiology frequently blends approaches: agent-based simulations capture rich behavioral detail, while differential equations support inference and policy optimization. Cellular automata occupy a productive middle ground. They are simpler than full agent-based models but retain explicit spatial representation, making them useful as fast prototypes for hypothesis generation and for communicating mechanisms to multidisciplinary audiences.

Algorithmic Foundations

Our model is defined on a two-dimensional grid where each cell represents one individual and each individual can be in one of four epidemiological states per generation:

- **0 - Susceptible (S):** healthy and able to acquire infection.
- **1 - Infected (I):** currently infectious and able to transmit to neighbors.
- **2 - Recovered/Immune (R):** temporarily protected after infection.
- **3 - Dead (D):** removed permanently (absorbing state).

The transition rules are applied simultaneously to all cells, producing a stochastic, spatial SIR(S) dynamic:

1. **S $\rightarrow$ I (local transmission):** a susceptible cell becomes infected with probability $p_{inf} = 1 - (1 - \beta)^n$, where n is the number of infected neighbors in the Moore neighborhood and β is the per-neighbor transmissibility.
2. **I $\rightarrow$ R (recovery):** an infected cell recovers with probability γ per time step.
3. **I $\rightarrow$ D (mortality):** an infected cell dies with probability μ per time step, reflecting disease severity; death is evaluated before recovery by default.
4. **R $\rightarrow$ S (waning immunity, optional):** a recovered cell returns to susceptible with probability ω, enabling recurrent seasonal-like dynamics when enabled.

Aligned with the provided Game of Life template, the Python implementation uses NumPy arrays to represent the lattice and Matplotlib to animate the evolution. The core algorithm consists of the following key functions:

- **criar_grade(linhas, colunas, frac_infectados)**: Initializes the grid as fully susceptible and seeds a fraction of infected individuals.
- **contar_vizinhos_infectados(grade, linha, coluna)**: Counts infected neighbors in the 8-neighborhood (Moore) while enforcing fixed boundaries.
- **atualizar_grade(grade, beta, gamma, mu, omega)**: Applies the stochastic transition rules synchronously, producing the next generation of states.
- **animar(frame, grade, img, params)**: Updates the grid per frame and redraws the image for an animated view of outbreak spread.

The infection probability function p_inf = 1 - (1 - beta)^n implements a simple independence assumption: each infected neighbor generates an exposure event with probability beta, and the susceptible individual becomes infected if at least one exposure succeeds. This form ensures that risk increases monotonically with the number of infected neighbors while remaining bounded in [0, 1]. It also provides a convenient knob for exploring the effect of repeated local exposures without requiring explicit contact tracing.

Time in the model is discretized into generations. For ILI, one generation can be interpreted as a fraction of a day or a day, depending on the calibration of gamma. Because we aim for exploratory analysis rather than exact fitting, parameters are chosen to produce qualitatively plausible dynamics: short infectious periods (moderate gamma), appreciable local transmissibility (beta), and low mortality (mu). Importantly, these parameters can be adjusted interactively to represent higher-risk variants or different respiratory pathogens.

Boundary conditions influence spatial dynamics. The implementation uses fixed boundaries (no wrap-around), which approximates a finite community. Periodic boundaries can be implemented by allowing indices to wrap, representing an effectively infinite tiling of the community. While boundary choices can affect edge behavior, the qualitative mechanisms of clustering and wavefront spread remain robust for sufficiently large grids.

Although the update step is written with explicit loops to mirror the didactic structure of the base code, the model remains computationally efficient for classroom-scale lattices (e.g., 50x50 to 150x150). For research-scale simulations, the same logic can be accelerated by vectorizing neighborhood counts (e.g., via convolution) and by batching random draws.

The parameterization is intentionally interpretable. Increasing β intensifies local spread and produces faster, more contiguous wavefronts; increasing γ shortens infectious periods and reduces peak prevalence; increasing μ increases the fraction of removed individuals and can suppress onward transmission by rapidly depleting the infectious pool. When $\omega > 0$, the system can exhibit recurrent flare-ups if susceptible pockets re-emerge near infectious clusters.

Applications and Impact

The primary contribution of this CA model is its ability to expose spatial mechanisms that are invisible in well-mixed models. When transmission is local, infections tend to cluster, leaving behind patchy landscapes of recovered and susceptible individuals. This patchiness can slow down spread, create irregular epidemic fronts, and generate long tails in incidence. Such effects help explain why identical average transmissibility can lead to different outbreak outcomes in communities with different densities or contact patterns.

From a public health perspective, the model can be used to prototype intervention logic. A reduction in contact intensity can be represented by lowering β globally (masking, improved ventilation) or locally (zoning, micro-lockdowns). Isolation of symptomatic cases can be approximated by reducing the infectious neighborhood influence for selected cells. Vaccination campaigns can be represented by initializing a fraction of the grid in the recovered/immune state. Even without explicit mobility, these simple changes can qualitatively reproduce expected outcomes: delayed peaks, smaller final size, and disrupted spatial connectivity of transmission chains.

Educationally, animated CA simulations are particularly effective for communicating non-intuitive concepts such as outbreak thresholds and the role of clustering. Students can observe that a small number of initial infections may die out stochastically even when average conditions favor spread, and that large outbreaks often require contiguous pathways of susceptible individuals. The model thus serves as a sandbox for building intuition before moving to parameter estimation, inference, and policy analysis using more complex frameworks.

Beyond qualitative visualization, the same simulation can be instrumented to produce standard epidemiological summary measures. By counting the number of cells in each state at every generation, one obtains time series analogous to $S(t)$, $I(t)$, $R(t)$, and $D(t)$. These can be used to compute peak prevalence, final epidemic size, and approximate growth rates. Because the model is stochastic, repeating runs under identical parameters yields a distribution of outcomes, enabling uncertainty-aware comparisons between intervention scenarios.

The CA representation is also well-suited for introducing heterogeneity. Spatially varying beta can represent higher-risk microenvironments (crowded indoor settings) embedded within a lower-risk background. Similarly, gamma and mu can vary to approximate risk gradients due to age, comorbidities, or healthcare access. Such heterogeneity often produces non-uniform burden across the lattice, highlighting inequities that are averaged out in well-mixed models.

Finally, the model provides a concrete platform for discussing data and inference. Although the present code is not calibrated to surveillance data, students can explore how observation processes (e.g., under-reporting of asymptomatic infections) would map from the latent I state to reported cases. This connection motivates why mechanistic simulation and statistical inference are complementary in epidemiological decision-making.

Conclusion

We presented a stochastic, spatial SIR(S) cellular automaton for an influenza-like illness, implemented in Python using the same structural approach as a classic cellular automaton demonstration. By combining local transmission with recovery, mortality, and optional waning immunity, the simulator captures key qualitative features of respiratory outbreaks, including clustering and wavefront spread.

Future extensions may incorporate explicit mobility, age-structured risk, heterogeneous susceptibility, or contact networks derived from real geospatial data. Coupling the simulator to data assimilation or agent-based calibration would enable more quantitative studies, while preserving the clarity and teachability that make cellular automata a powerful first-principles modeling tool.

References

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